2021 Taiwan IMO Camp — A5

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Let M be an arbitrary positive real number greater than 1 , and let $a_1, a_2, \ldots$ be an infinite sequence of real numbers with $a_n \in [1, M]$ for any $n \in \mathbb{N}$. Show that for any $\varepsilon > 0$, there exists a positive integer n such that

$$\frac{a_n}{a_{n+1}} + \frac{a_{n+1}}{a_{n+2}} + \cdots + \frac{a_{n+t-1}}{a_{n+t}} \geq t - \varepsilon$$

holds for any positive integer t.

Solution

If we can find n such that $a_n \geq a_{n+1} \geq \cdots \geq a_{n+t}$ then we will be done because that sum on the left hand side will be $\geq t$. Hence assume, for the sake of contradiction, that such n cannot be found.

If we cannot find such n, then it must be the case that (a_n) is strictly increasing. Because (a_n) is bounded and strictly increasing, we can deduce that $\lim_{n\to\infty} a_n = M$ and so $\lim_{n\to\infty}(a_n - a_{n-1}) = 0$, and thus we can say the following: for some sufficiently large integer N, it is possible to find a contiguous sub-sequence $a_N, a_{N+1}, \ldots, a_{N+t}$ with the following property

$$a_{N+k} + \varepsilon_{N+k} = a_{N+k+1}$$

for integers $0 \leq k < t$, and for some *very* small positive (ε_n). Now we observe that

$$\frac{a_N}{a_{N+1}} + \frac{a_{N+1}}{a_{N+2}} + \cdots + \frac{a_{N+t-1}}{a_{N+t}} = t - \left(\frac{\varepsilon_N}{a_N} + \frac{\varepsilon_{N+1}}{a_{N+1}} + \cdots + \frac{\varepsilon_{N+t}}{a_{N+t}}\right)$$

Now letting $\varepsilon = \left(\frac{\varepsilon_N}{a_N} + \frac{\varepsilon_{N+1}}{a_{N+1}} + \cdots + \frac{\varepsilon_{N+t}}{a_{N+t}}\right)$ finishes the problem.